TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Senior Project I

TEDU Guide — Project Backlog v2

Spring 2026

Project Name: TEDU Guide

Project Website: https://tedu-guide.vercel.app/

Team Members

1. Taci Furkan Gökalp
2. Elif Nazlı Özer
3. Fatima Nasir Muhammad

Advisor & Jury Members

Advisor: Tolga Kurtuluş Çapın

Jury Member 1: Mehmet Evren Çoşkun

Jury Member 2: Müslüm Bozyiğit

Backlog Overview

The backlog is organized into five phases aligned with the project timeline. Each item is tagged with:

- **Task ID**: Unique task identifier
- **Priority**: Critical / High / Medium / Low
- **Status**: Not Started / In Progress / Completed / Blocked
- **Estimated Time**: Linked FR / NFR / PR from the Specification Report
- **Assigned To**: Team member responsible

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1. Project Phases & Timeline Overview

Phase	Title	Duration	Target Completion	Status
Phase 0	Project Setup & Planning	Week 1 – Week 2	27-Mar-26	In Progress
Phase 1	Data Collection & Infrastructure	Week 3 – Week 5	17-Apr-26	Not Started
Phase 2	Core Feature Development	Week 6 – Week 10	22-May-26	Not Started
Phase 3	Compliance & Non-Functional	Week 9 – Week 11	22-May-26	Not Started
Phase 4	Integration & Final Demo	Week 11 – Week 13	12-Jun-26	Not Started

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2. Detailed Task Backlog

Phase 0: Planning & Setup

ID	Task Description	Priority	Assigned To	Status
P0-01	Project Website Deployment (Vercel)	Critical	Taci Furkan	Completed
P0-02	D1: Project Proposal Submission	Critical	All Team	Completed
P0-	D2: Project Specifications &	Critical	All Team	Completed

ID	Task Description	Priority	Assigned To	Status
03	Literature			
P0-04	D3: Initial Backlog Document (v1)	Critical	Taci Furkan	Completed
P0-05	D4: Project Analysis Report	Critical	All Team	Completed
P0-06	D5: Backlog Document (v2)	Critical	Taci Furkan	In Progress
P0-08	Formal IT Approval for WiFi Data Collection	Critical	Fatima Nasir	Not Started

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Phase 1: Data & Infrastructure

ID	Task Description	Priority	Assigned To	Requirement
P1-01	Digitize floor plans (Leaflet/SVG format)	Critical	Elif Nazlı	FR-06
P1-03	WiFi RSSI Fingerprinting Site Survey	Critical	Fatima Nasir	FR-01
P1-06	PostGIS Spatial Database Schema Design	High	Taci Furkan	PR-01
P1-08	React Frontend + Leaflet.js Scaffold	High	Taci Furkan	PR-04

Phase 2: Core Development (AI & Navigation)

ID	Task Description	Priority	Assigned To	Requirement
P2A-02	Train k-NN Indoor Positioning Model	Critical	Fatima Nasir	FR-02
P2B-02	OpenAI API Integration (FastAPI)	Critical	Elif Nazlı	FR-03
P2C-02	A* / Dijkstra Shortest Path Implementation	Critical	Taci Furkan	FR-04
P2B-06	Speech-to-Text Voice Input Integration	Medium	Fatima Nasir	FR-09

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3. Deliverable Checklist

Deliverable	Due Date	Status	Format
D1 — Project Proposal	20-Mar-26	Completed	PDF (Email)
D2 — Specification Report	27-Mar-26	Completed	PDF (Email + LMS)
D3 — Backlog v1	27-Mar-26	Completed	PDF (Email + LMS)
D4 — Analysis Report	10-Apr-26	Completed	DOCX (LMS + Website)
D5 — Backlog v2	10-Apr-26	In Progress	PDF (LMS)
D6 — High-Level Design Report	15-May-26	Not Started	PDF (LMS + Website)
D7 — Backlog v3	15-May-26	Not Started	PDF (LMS)
D8 — Final Presentation	TBA	Not Started	Oral + Demo

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4. Risk & Mitigation Table

ID	Risk Description	Impact	Mitigation Strategy
R01	WiFi Signal Interference / Noise	High	Use signal smoothing and dense fingerprint grids.
R02	IT Department Approval Delays	Critical	Submit request immediately; use manual pin-drop fallback.
R03	API Cost Constraints (OpenAI)	Medium	Optimize prompts; use GPT-3.5 for dev, GPT-4 for production.
R04	Floor Plan Availability	High	Coordinate with university administration early.
R05	Team Availability During Finals	Medium	Front-load critical tasks before Week 11.